PH.30102 **FALL 2026**

Quantum Mechanics III

Homework 4

Seonwoo Park Due 2026-09-27 23:59

Topic

Open Systems and Lindblad Generators

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Prove the first displayed relation used in Open Systems and Lindblad Generators. State all assumptions and conventions.

$$\dot{\rho} = -i[H, \rho] + \sum_j \gamma_j (L_j \rho L_j^\dagger - \frac{1}{2} L_j^\dagger L_j, \rho)$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$T_1^{-1} = \gamma_\downarrow + \gamma_\uparrow$$

3. Interpret the third relation in two limiting regimes and explain the physical meaning of the result.

$$T_2^{-1} = 1/(2T_1) + \gamma_\varphi$$